

VISUALIZING US NATURAL DIASTER DECLARATION – TRENDS AND PATTERNS

Week 1 Documentation/ Screenshots

Data Cleaning in Power BI

Data Cleaning in Power BI

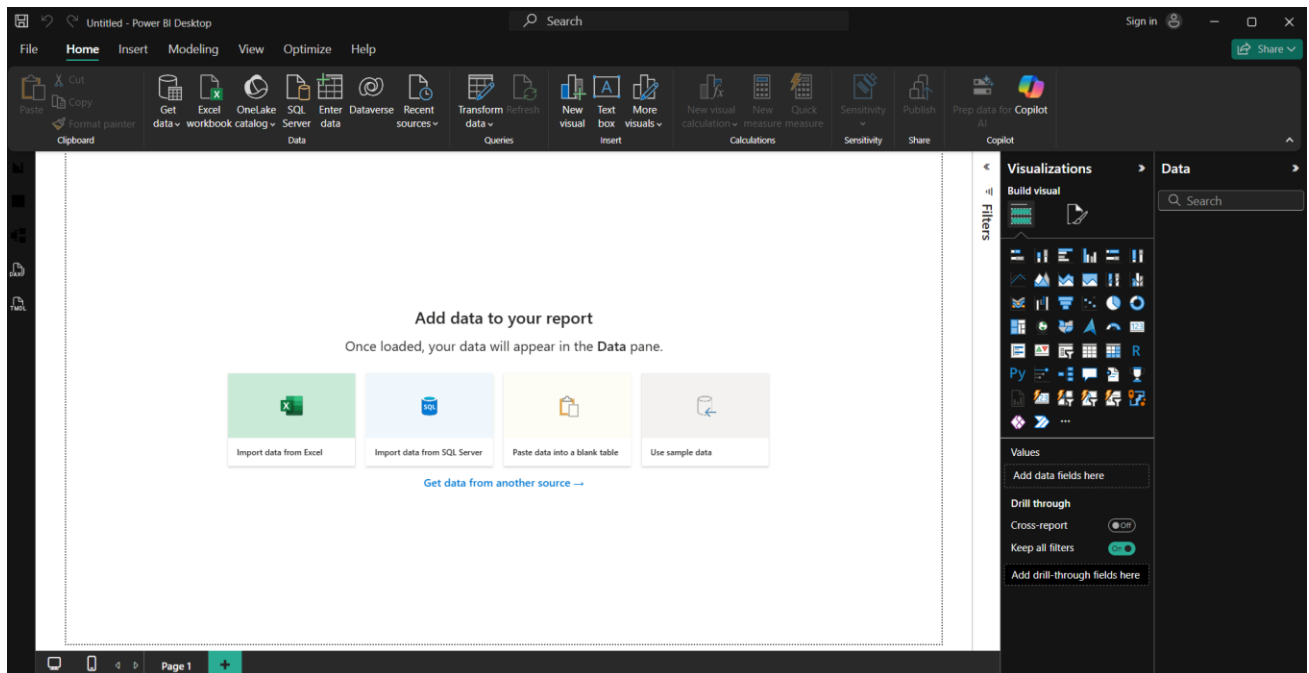


Fig 1.1 PowerBI Canvas

Load Data:

Get Data -> Text/CSV -> Choose File -> Transform Data

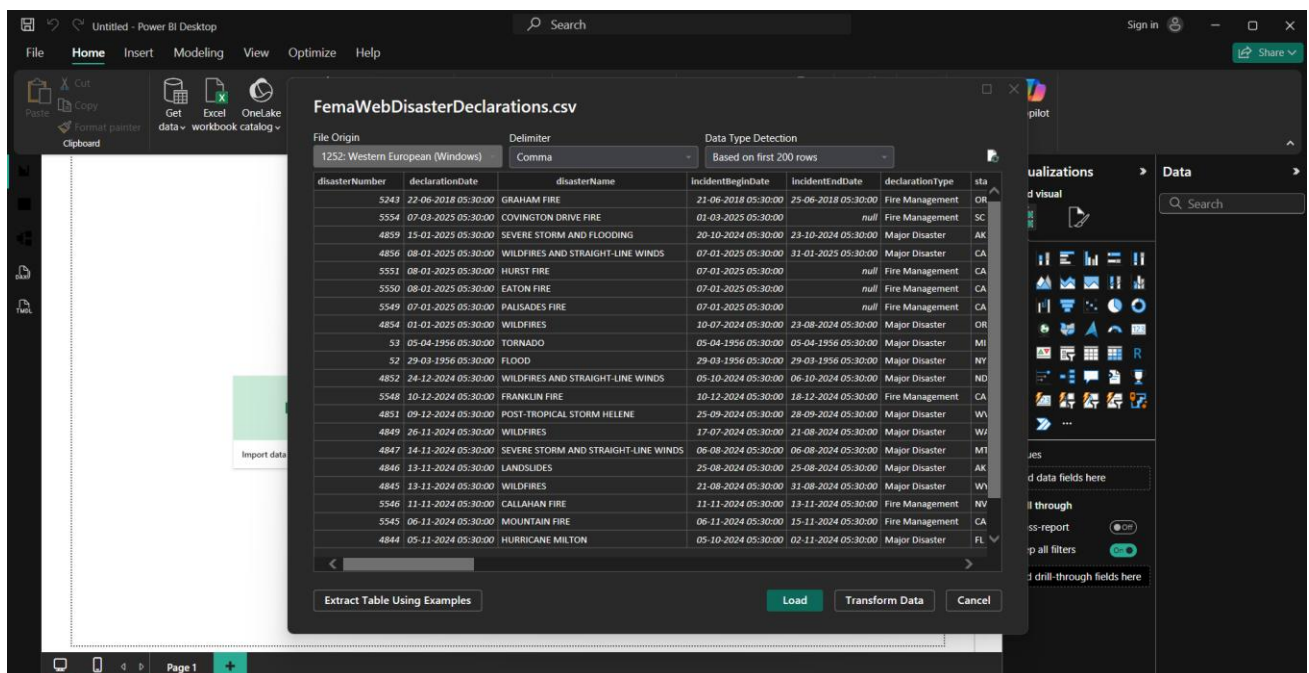


Fig 1.2 Transform Data

Remove Duplicate Rows:

Ctrl+A -> Remove Rows -> Remove Duplicates

Remove Unnecessary Columns:

Click StateCode column, Ctrl+alt+(other columns to remove) -> Remove Columns (in toolbar ribbon) -> Remove Columns

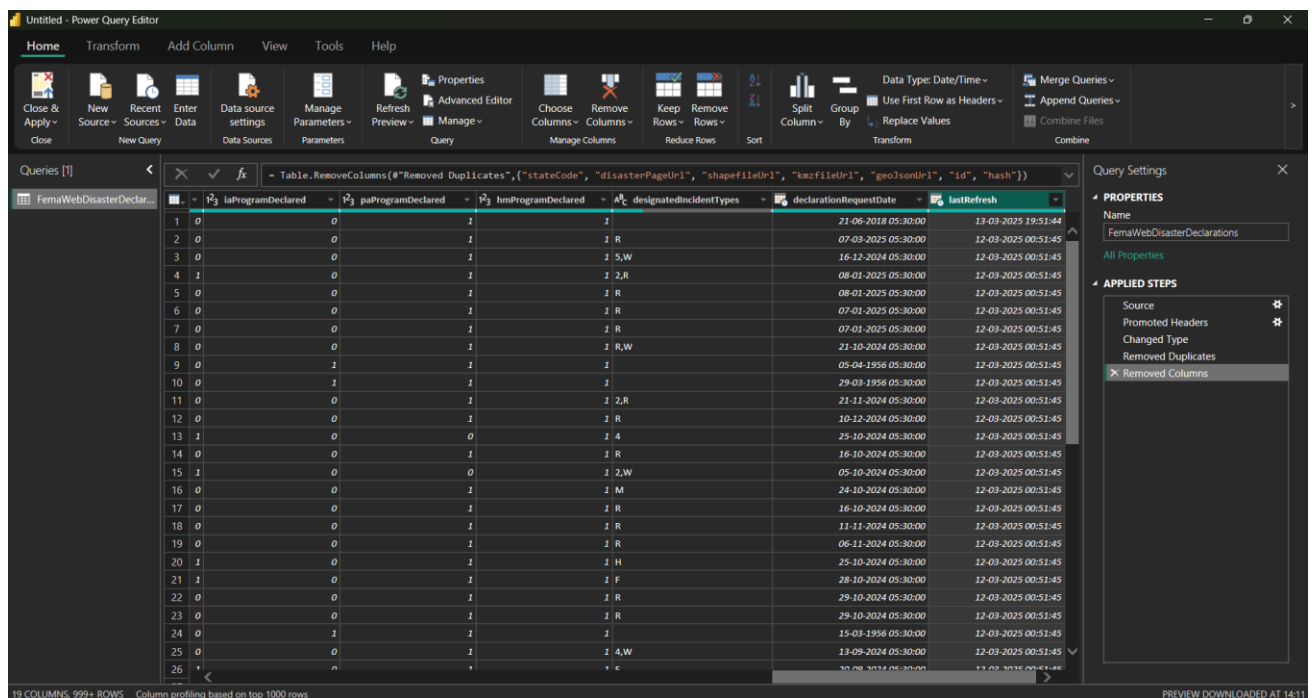


Fig 1.3 Removed unnecessary columns

Check Datatype:

To convert disaster programs declared to Boolean: Click the number icon -> true/false

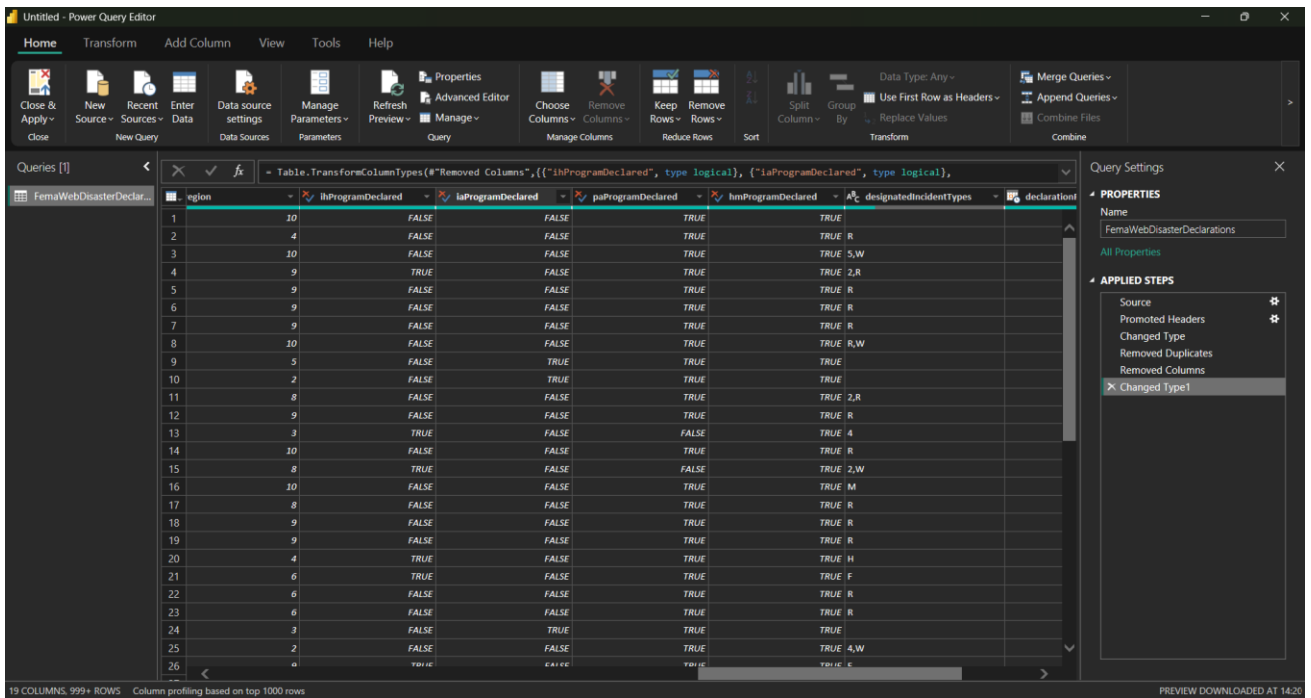


Fig 1.4 Changed Datatype

Replace blank with Unknown: (in designatedIncidentType)

Select column -> replace values -> , Unknown

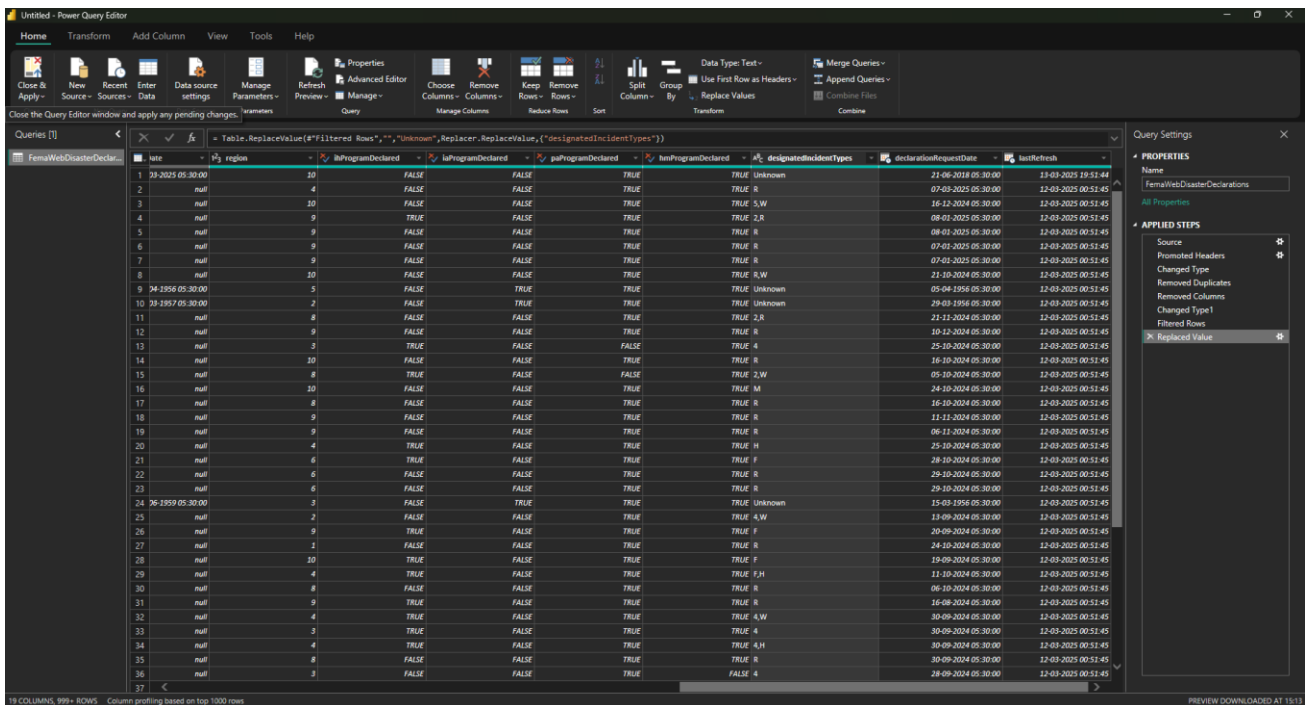


Fig 1.5 Replaced blank values

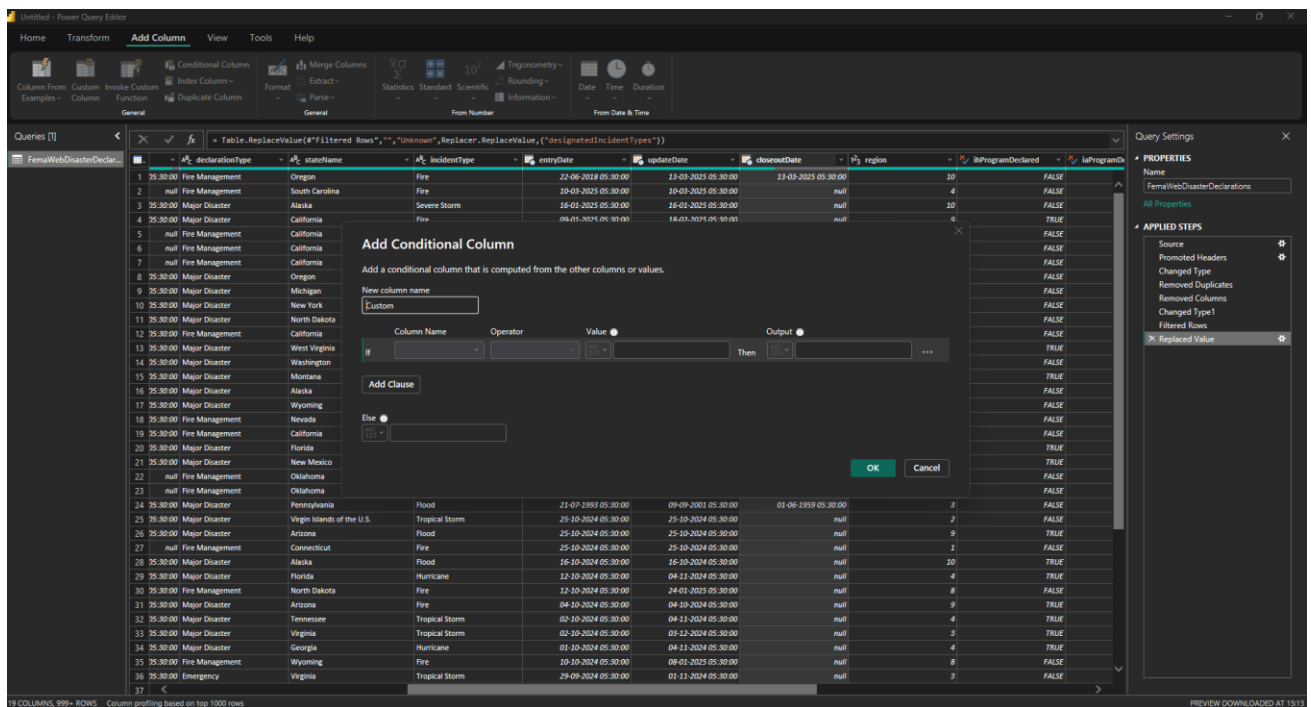


Fig 1.7 “Add Conditional Column” pop-up

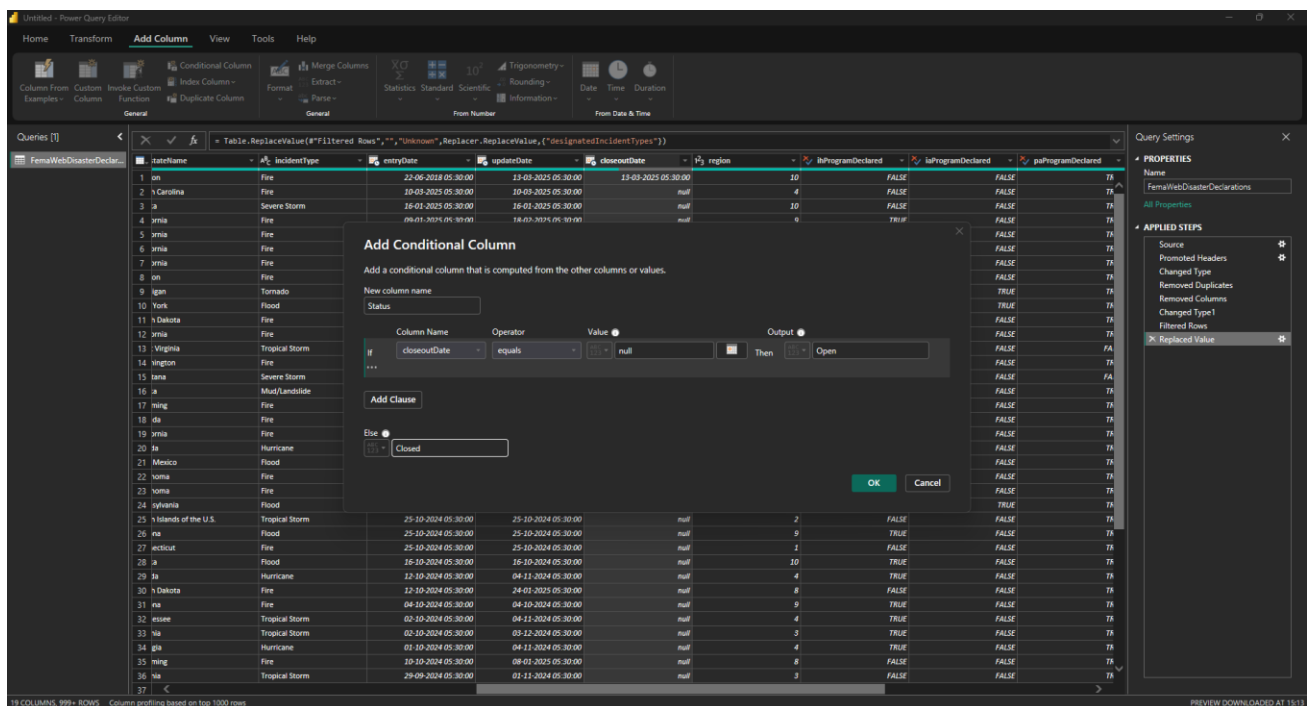


Fig 1.8 “Add Conditional Column” pop-up

Figure 1.9 shows a screenshot of the Power Query Editor interface. The main area displays a table with 37 rows and 9 columns. The columns are: id, entryDate, updateDate, closeoutDate, Status, region, ipProgramDeclared, iaProgramDeclared, and paProgramDeclared. The 'Status' column is highlighted in blue. The right-hand pane shows the 'Query Settings' for 'FemaWebDisasterDeclarations', with the 'APPLIED STEPS' list containing 'Reordered Columns'.

Fig 1.9 Created a conditional column “Status”

2. Incident Duration

= Duration.Days([incidentEndDate] - [incidentBeginDate])

In header, Add Column -> Custom Column

Figure 1.10 shows a screenshot of the Power Query Editor interface with the 'Custom Column' pop-up dialog open. The dialog has a text box for 'New column name' and a text box for 'Custom column formula'. The formula box contains the text: '= Duration.Days([incidentEndDate] - [incidentBeginDate])'. The 'Available columns' list on the right includes: disasterNumber, declarationDate, disasterName, incidentBeginDate, incidentEndDate, declarationType, and stateName. The 'Main' area shows a table with 37 rows and 9 columns, including a new column 'IncidentDuration'.

Fig 1.10 Custom Column pop-up

IncidentName	IncidentBeginDate	IncidentEndDate	incidentDuration	A1 declarationType	A1 stateName	A1 incidentType	entryDate	updateDate
1 SAM FIRE	21-06-2018 05:30:00	25-06-2018 05:30:00	4	Fire Management	Oregon	Fire	22-06-2018 05:30:00	15-03
2 NGTON DRIVE FIRE	01-03-2025 05:30:00	null	null	Fire Management	South Carolina	Fire	10-03-2025 05:30:00	10-03
3 RE STORM AND FLOODING	20-10-2024 05:30:00	23-10-2024 05:30:00	3	Major Disaster	Alaska	Severe Storm	16-01-2025 05:30:00	15-01
4 HRES AND STRAIGHT-LINE WINDS	07-01-2025 05:30:00	31-01-2025 05:30:00	24	Major Disaster	California	Fire	08-01-2025 05:30:00	18-01
5 JT FIRE	07-01-2025 05:30:00	null	null	Fire Management	California	Fire	08-01-2025 05:30:00	08-01
6 IN FIRE	07-01-2025 05:30:00	null	null	Fire Management	California	Fire	08-01-2025 05:30:00	08-01
7 JADES FIRE	07-01-2025 05:30:00	null	null	Fire Management	California	Fire	08-01-2025 05:30:00	08-01
8 HRES	10-07-2024 05:30:00	23-08-2024 05:30:00	44	Major Disaster	Oregon	Fire	02-01-2025 05:30:00	02-01
9 AAO	05-04-1956 05:30:00	05-04-1956 05:30:00	0	Major Disaster	Michigan	Tornado	21-07-1993 05:30:00	09-09
10 ID	29-03-1956 05:30:00	29-03-1956 05:30:00	0	Major Disaster	New York	Flood	21-07-1993 05:30:00	09-09
11 HRES AND STRAIGHT-LINE WINDS	05-10-2024 05:30:00	06-10-2024 05:30:00	1	Major Disaster	North Dakota	Fire	25-12-2024 05:30:00	25-12
12 KLIN FIRE	10-11-2024 05:30:00	18-11-2024 05:30:00	8	Fire Management	California	Fire	10-11-2024 05:30:00	07-02
13 TROPICAL STORM HELENE	25-09-2024 05:30:00	28-09-2024 05:30:00	3	Major Disaster	West Virginia	Tropical Storm	09-12-2024 05:30:00	09-12
14 HRES	17-07-2024 05:30:00	21-08-2024 05:30:00	35	Major Disaster	Washington	Fire	27-11-2024 05:30:00	27-11
15 RE STORM AND STRAIGHT-LINE WL	06-08-2024 05:30:00	06-08-2024 05:30:00	0	Major Disaster	Montana	Severe Storm	14-11-2024 05:30:00	14-11
16 YLDES	25-08-2024 05:30:00	25-08-2024 05:30:00	0	Major Disaster	Alaska	Mud/Landslide	14-11-2024 05:30:00	14-11
17 HRES	21-08-2024 05:30:00	31-08-2024 05:30:00	10	Major Disaster	Wyoming	Fire	14-11-2024 05:30:00	14-11
18 AHAN FIRE	11-11-2024 05:30:00	11-11-2024 05:30:00	2	Fire Management	Nevada	Fire	11-11-2024 05:30:00	07-02
19 INTAIN FIRE	06-11-2024 05:30:00	15-11-2024 05:30:00	9	Fire Management	California	Fire	12-11-2024 05:30:00	07-02
20 ICANE MILTON	05-10-2024 05:30:00	02-11-2024 05:30:00	28	Major Disaster	Florida	Hurricane	06-11-2024 05:30:00	20-11
21 RE STORM AND FLOODING	19-10-2024 05:30:00	20-10-2024 05:30:00	1	Major Disaster	New Mexico	Flood	01-11-2024 05:30:00	01-11
22 IN ROAD FIRE	20-10-2024 05:30:00	null	null	Fire Management	Oklahoma	Fire	30-10-2024 05:30:00	30-10
23 REC CREEK FIRE	20-10-2024 05:30:00	null	null	Fire Management	Oklahoma	Fire	30-10-2024 05:30:00	30-10
24 ID	15-03-1956 05:30:00	15-03-1956 05:30:00	0	Major Disaster	Pennsylvania	Flood	21-07-1993 05:30:00	09-09
25 NCAL STORM ERNESTO	13-08-2024 05:30:00	16-08-2024 05:30:00	3	Major Disaster	Virgin Islands of the U.S.	Tropical Storm	25-10-2024 05:30:00	25-10
26 IDING	22-08-2024 05:30:00	23-08-2024 05:30:00	1	Major Disaster	Arizona	Flood	25-10-2024 05:30:00	25-10
27 THORNE FIRE	21-10-2024 05:30:00	null	null	Fire Management	Connecticut	Fire	25-10-2024 05:30:00	25-10
28 IDING	05-08-2024 05:30:00	06-08-2024 05:30:00	1	Major Disaster	Alaska	Flood	16-10-2024 05:30:00	16-10
29 ICANE MILTON	05-10-2024 05:30:00	02-11-2024 05:30:00	28	Major Disaster	Florida	Hurricane	12-10-2024 05:30:00	04-11
30 IDEN FIRE	05-10-2024 05:30:00	23-10-2024 05:30:00	16	Fire Management	North Dakota	Fire	12-10-2024 05:30:00	24-01
31 CH FIRE	10-07-2024 05:30:00	17-07-2024 05:30:00	7	Major Disaster	Arizona	Fire	04-10-2024 05:30:00	04-10
32 NCAL STORM HELENE	26-09-2024 05:30:00	30-09-2024 05:30:00	4	Major Disaster	Tennessee	Tropical Storm	02-10-2024 05:30:00	04-11
33 NCAL STORM HELENE	25-09-2024 05:30:00	03-10-2024 05:30:00	8	Major Disaster	Virginia	Tropical Storm	02-10-2024 05:30:00	03-12
34 ICANE HELENE	24-09-2024 05:30:00	30-10-2024 05:30:00	36	Major Disaster	Georgia	Hurricane	01-10-2024 05:30:00	04-11
35 IRE	28-09-2024 05:30:00	22-10-2024 05:30:00	24	Fire Management	Wyoming	Fire	10-10-2024 05:30:00	08-01
36 TROPICAL CYCLONE HELENE	25-09-2024 05:30:00	03-10-2024 05:30:00	8	Emergency	Virginia	Tropical Storm	29-09-2024 05:30:00	01-11

Fig 1.11 Added “incidentDuration” column

Change datatype of “incidentDuration” – whole number

3. Fiscal Year

In header, Add Column -> Custom Column

= if (Date.Month ([declarationDate]) >= 10)
 then (Date.Year([declarationDate]) + 1)
 else (Date.Year([declarationDate]))

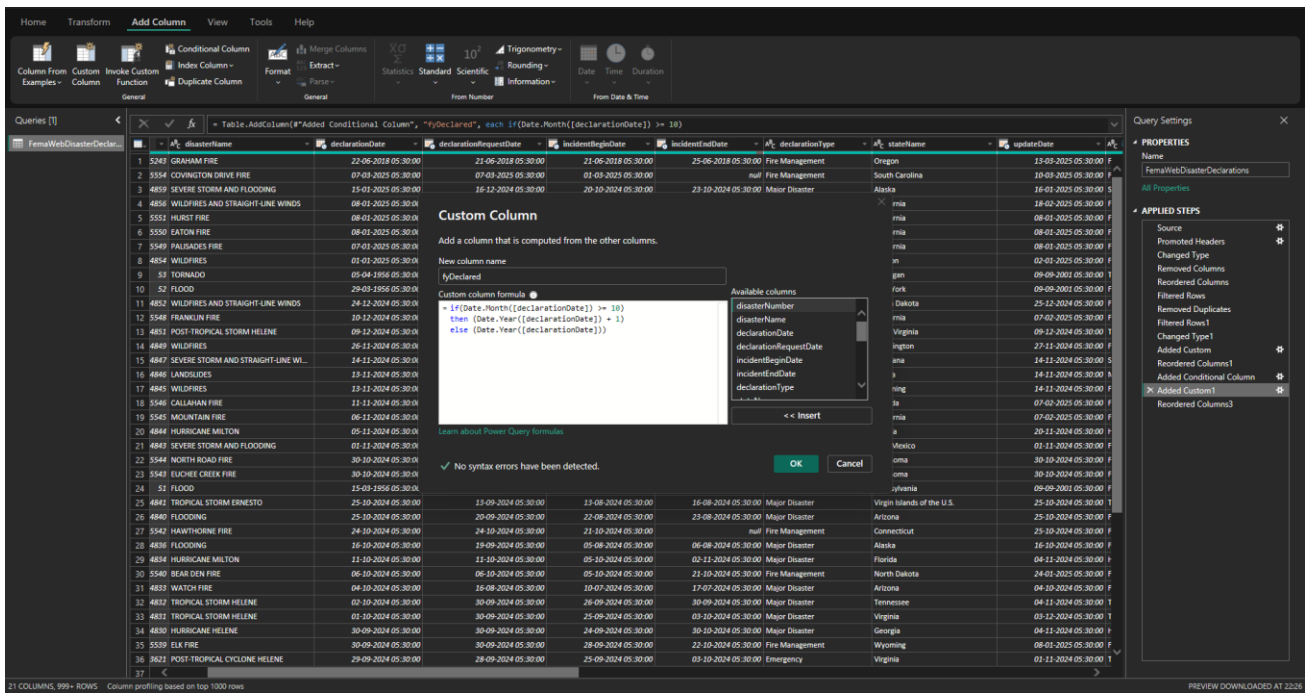


Fig 1.12 Custom Column pop-up for “fyDeclared” column

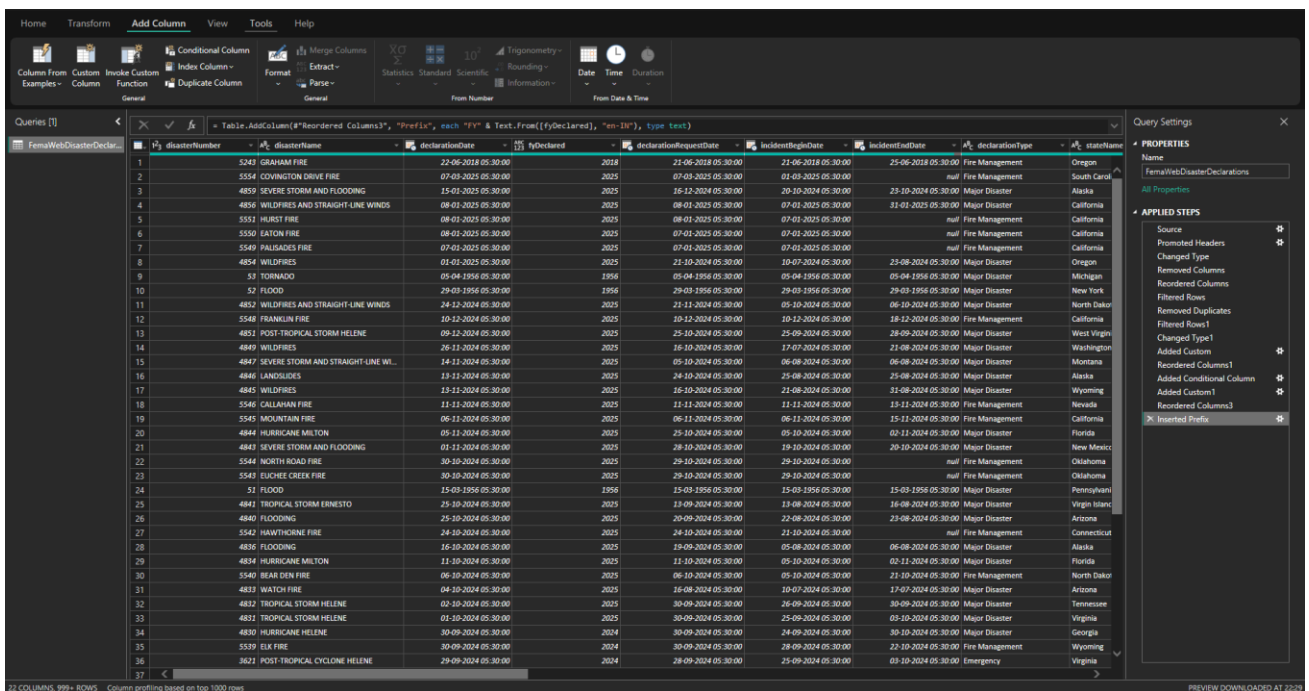


Fig 1.13 Added fyDeclared column

Select column -> Format -> Add Prefix -> FY

Delete previous column

Rename column

Home Transform **Add Column** View Tools Help

Column From Examples Custom Invoke Custom Function Duplicate Column

General

Format Extract Parse

Statistics Standard Scientific Rounding Information

From Number

Date Time Duration

From Date & Time

Queries [1]

FormulaWebDisasterDeclarations

Table.ReorderColumns({"Renamed Columns", "disasterNumber", "disasterName", "declarationDate", "fyDeclared", "declarationRequestDate", "incidentBeginDate", "incidentEndDate", "declarationType", "stateName", "updateDate"})

	Name	declarationDate	fyDeclared	declarationRequestDate	incidentBeginDate	incidentEndDate	declarationType	stateName	updateDate
2	I DRIVE FIRE	07-09-2025 05:30:00	FY2025	07-09-2025 05:30:00	01-03-2025 05:30:00	null	Fire Management	South Carolina	10-09-20
3	RM AND FLOODING	15-01-2025 05:30:00	FY2025	16-12-2024 05:30:00	20-10-2024 05:30:00	23-10-2024 05:30:00	Major Disaster	Alaska	16-02-20
4	IND STRAIGHT-LINE WINDS	08-01-2025 05:30:00	FY2025	08-01-2025 05:30:00	07-01-2025 05:30:00	31-01-2025 05:30:00	Major Disaster	California	18-02-20
5		08-01-2025 05:30:00	FY2025	08-01-2025 05:30:00	07-01-2025 05:30:00	null	Fire Management	California	08-03-20
6		08-01-2025 05:30:00	FY2025	07-01-2025 05:30:00	07-01-2025 05:30:00	null	Fire Management	California	08-03-20
7	IRE	07-09-2025 05:30:00	FY2025	07-09-2025 05:30:00	07-01-2025 05:30:00	23-08-2024 05:30:00	Major Disaster	Oregon	02-02-20
8		01-09-2025 05:30:00	FY2025	21-10-2024 05:30:00	10-07-2024 05:30:00	05-04-1956 05:30:00	Major Disaster	Michigan	09-09-20
9		05-04-1956 05:30:00	FY1956	05-04-1956 05:30:00	05-04-1956 05:30:00	05-04-1956 05:30:00	Major Disaster	New York	09-09-20
10		29-09-1956 05:30:00	FY1956	29-09-1956 05:30:00	29-09-1956 05:30:00	29-09-1956 05:30:00	Major Disaster	North Dakota	25-12-20
11	IND STRAIGHT-LINE WINDS	24-12-2024 05:30:00	FY2025	21-11-2024 05:30:00	05-10-2024 05:30:00	06-10-2024 05:30:00	Major Disaster	California	07-02-20
12	IRE	10-12-2024 05:30:00	FY2025	10-12-2024 05:30:00	10-12-2024 05:30:00	18-12-2024 05:30:00	Fire Management	West Virginia	09-12-20
13	CAL STORM HELENE	09-12-2024 05:30:00	FY2025	25-10-2024 05:30:00	25-09-2024 05:30:00	28-09-2024 05:30:00	Major Disaster	Washington	27-12-20
14		26-11-2024 05:30:00	FY2025	16-10-2024 05:30:00	17-07-2024 05:30:00	21-08-2024 05:30:00	Major Disaster	Montana	14-11-20
15	RM AND STRAIGHT-LINE WINDS	14-11-2024 05:30:00	FY2025	09-10-2024 05:30:00	06-08-2024 05:30:00	06-08-2024 05:30:00	Major Disaster	Alaska	14-11-20
16		13-11-2024 05:30:00	FY2025	24-10-2024 05:30:00	25-08-2024 05:30:00	25-08-2024 05:30:00	Major Disaster	Wyoming	14-11-20
17		13-11-2024 05:30:00	FY2025	16-10-2024 05:30:00	21-08-2024 05:30:00	31-08-2024 05:30:00	Major Disaster	Nevada	07-02-20
18	IRE	11-11-2024 05:30:00	FY2025	11-11-2024 05:30:00	11-11-2024 05:30:00	13-11-2024 05:30:00	Fire Management	California	07-02-20
19	FIRE	06-11-2024 05:30:00	FY2025	06-11-2024 05:30:00	06-11-2024 05:30:00	15-11-2024 05:30:00	Fire Management	Florida	20-12-20
20	MILTON	05-11-2024 05:30:00	FY2025	25-10-2024 05:30:00	05-10-2024 05:30:00	02-11-2024 05:30:00	Major Disaster	New Mexico	01-11-20
21	RM AND FLOODING	01-11-2024 05:30:00	FY2025	28-10-2024 05:30:00	19-10-2024 05:30:00	20-10-2024 05:30:00	Major Disaster	Oklahoma	10-10-20
22	D FIRE	30-10-2024 05:30:00	FY2025	29-10-2024 05:30:00	29-10-2024 05:30:00	null	Fire Management	Oklahoma	10-10-20
23	EK FIRE	30-10-2024 05:30:00	FY2025	29-10-2024 05:30:00	29-10-2024 05:30:00	null	Fire Management	Pennsylvania	09-09-20
24		15-09-1956 05:30:00	FY1956	15-09-1956 05:30:00	15-03-1956 05:30:00	15-03-1956 05:30:00	Major Disaster	Virgin Islands of the U.S.	25-10-20
25	FORM ERNESTO	25-10-2024 05:30:00	FY2025	13-09-2024 05:30:00	13-08-2024 05:30:00	16-08-2024 05:30:00	Major Disaster	Arizona	25-10-20
26		25-10-2024 05:30:00	FY2025	20-09-2024 05:30:00	22-08-2024 05:30:00	23-08-2024 05:30:00	Major Disaster	Connecticut	16-10-20
27	E FIRE	24-10-2024 05:30:00	FY2025	24-10-2024 05:30:00	21-10-2024 05:30:00	null	Fire Management	Florida	04-11-20
28		16-10-2024 05:30:00	FY2025	19-09-2024 05:30:00	05-08-2024 05:30:00	06-08-2024 05:30:00	Major Disaster	North Dakota	24-02-20
29	MILTON	11-10-2024 05:30:00	FY2025	11-10-2024 05:30:00	05-10-2024 05:30:00	02-11-2024 05:30:00	Major Disaster	Arizona	04-10-20
30	IRE	06-10-2024 05:30:00	FY2025	06-10-2024 05:30:00	05-10-2024 05:30:00	21-10-2024 05:30:00	Fire Management	Tennessee	03-12-20
31		04-10-2024 05:30:00	FY2025	16-08-2024 05:30:00	10-07-2024 05:30:00	17-07-2024 05:30:00	Major Disaster	Georgia	08-11-20
32	FORM HELENE	02-10-2024 05:30:00	FY2025	30-09-2024 05:30:00	26-09-2024 05:30:00	30-09-2024 05:30:00	Major Disaster	Wyoming	08-02-20
33	FORM HELENE	01-10-2024 05:30:00	FY2025	30-09-2024 05:30:00	25-09-2024 05:30:00	03-10-2024 05:30:00	Major Disaster	Virginia	01-11-20
34	HELENE	30-09-2024 05:30:00	FY2024	30-09-2024 05:30:00	24-09-2024 05:30:00	30-10-2024 05:30:00	Major Disaster	Emergency	01-11-20
35		30-09-2024 05:30:00	FY2024	30-09-2024 05:30:00	28-09-2024 05:30:00	23-10-2024 05:30:00	Fire Management	South Carolina	20-11-20
36	CAL CYCLONE HELENE	29-09-2024 05:30:00	FY2024	28-09-2024 05:30:00	25-09-2024 05:30:00	03-10-2024 05:30:00	Emergency		
37	FORM DEBBY	29-09-2024 05:30:00	FY2024	16-09-2024 05:30:00	04-08-2024 05:30:00	22-08-2024 05:30:00	Major Disaster		

21 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 22:31

Query Settings

PROPERTIES

Name

FormulaWebDisasterDeclarations

APPLIED STEPS

Source

Promoted Headers

Changed Type

Removed Columns

Reordered Columns

Filtered Rows

Removed Duplicates

Filtered Row1

Changed Type1

Added Custom

Reordered Columns1

Added Conditional Column

Added Custom1

Reordered Columns3

Inserted Prefix

Removed Columns1

Renamed Columns

Renamed Columns2

Fig 1.14 Final fyDeclared column